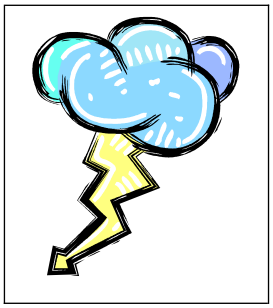
LESSON 2 E&C DANCING WITH STATIC



AIM OF THE LESSON

During this lesson students will:

- Describe how a transfer of energy from one material to another can create static electricity.
- Name some examples of static electricity.

EQUIPMENT

Per Class

- 1 foam cup filled with water
- 1 compass or sharp object
- 5 tea bags
- a variety of materials eg nylon; wool; fur; silk; paper; cotton; rubber; chux; stockings; polyester; plastic; sponge (a few of each is best)

Per Group

- 2 balloons
- 1 plastic comb or plastic pen cover
- 1 ball of wool (not acrylic) or piece of wool cloth
- 1 empty soft drink can
- 5 or 6 styrofoam balls/ packing foam/ beans out of bean bag or puffed wheat

Preparation

- Collection of materials to test (Students could bring in one each and could be a piece of clothing as they do not need to be cut).

Safety

- Nil



BACKGROUND INFORMATION FOR TEACHERS

Electricity

- The word *electricity* comes from the Greek word for amber – *electron.*
- Electricity is a form of energy that is used in many ways in daily life.
- Electricity can be observed in the natural environment – eg lightning.
- All matter is made of small particles called atoms.
- Within atoms are tiny particles called electrons which are negatively charged.

Static Electricity

- When materials come in contact with each other electrons may be transferred from one material to the other.
- This causes a build up of charge which is called static electricity.
- If a substance picks up electrons it becomes negatively charged.
- When materials have different charges they attract each other. When they have similar charges they repel.
- When there is a very big difference between these charges, when the objects come in contact they produce a spark.
- By rubbing certain objects together they become charged with electricity.
- Wool cloth and human hair readily give up electrons.
- Humid weather reduces the effectiveness of static electricity.
- When students rub materials on their hair it is best if the hair is fine and dry with no gel or hair spray.

Electric Shocks

- If you rub your feet on a carpet, your entire body can become charged.
- The excess electrons in your body need to go somewhere. Therefore, when you touch something that is not charged (like a large piece of metal) it absorbs the extra electrons suddenly.
- The electrons flow quickly through the tip of your finger into the object you touch with a ZAP!
- The flow of electrons might be great enough for you to see a tiny spark and hear a snapping sound.

Balloons

- A balloon rubbed with a wool cloth becomes negatively charged.



LESSON OUTLINE

1. Introduction (Engaging Students)

Demonstration

- Fill a foam cup with water.
- Ask a student to rub a plastic comb with a ball of wool or their hair for 10 secs but do not put the comb down.
- Use a compass to put a tiny hole in the bottom of the foam cup to produce a very fine stream of water (a tap is better if available).
- As soon as the water starts to flow hold the charged comb next to the stream of water without touching it.

What is happening?	The comb should cause the stream of water to bend towards it.
What is causing the water to bend towards the comb?	The comb is attracting it like a magnet. Transfer of energy or electrons.
Where have you *seen or felt* electricity?	Getting a shock when stepping out of a car. Combing hair causes unusual attractions. Turning the TV off sometimes causes a zap and shock. Lightning. Clothes out of the dryer cling together.

2. Activities and Strategies

- Write down the aim, equipment and method for the activity as a class.
- Discuss what is meant by a prediction. Students write a prediction for each activity.
- Students conduct the activities on the handout.

3. Focus Questions (Explain and Elaborate)

Why are some of the materials attracted to the comb?	Opposite charges.
Why are some of the materials pushed away or repelled by the comb?	Like charges.
What cause these different charges?	Electrons moving from one material to another.
How does static electricity travel compared to electricity in our homes	Static electricity is through the air. Other electricity travels in wires.

4. Reflection

- In groups, students share one idea that summarises what they have learnt about static electricity.
- These could then be shared with the class.
- Students to write a conclusion.



EXTENSION ACTIVITIES

- Investigate a range of other materials found in the classroom to see if they attract or repel other materials e.g. plastic rulers, erasers, pencil cases. Record your results in a table.



LINKS TO OTHER LEARNING AREAS

- Write a poem about lightning.
- Design a bookmark using lightning or static electricity as the focus.
- Research about Benjamin Franklin who flew a kite in a lightning storm.
- Write a story about a lightning storm.
- Make up a crossword puzzle or word sleuth that includes the types of materials you have used in this lesson.
- Find out if we see lightning or hear thunder first and why this is so.

ACTIVITY 2 DANCING WITH STATIC



Aim (What are we going to find out?)

Equipment (What do I need to complete the task?)

Method (How is the task to be done?)

Results (What did I observe?)

No	**Method**	**Prediction** (What I think will happen)	**Observations** (What I saw)
1	Rub a balloon with a ball of wool and hold the balloon over a friend's hair.		
2	Rub two balloons with a ball of wool and then hold the balloons side by side.		
3	Rub a balloon on your hair and hold it near some small torn up pieces of paper.		
4	Rub a balloon on a ball of wool, see if you can get it to stick to the wall.		
5	Rub a comb with a ball of wool and bring the balloon close to the foam balls. Wait to see what happens.		
6	Rub a comb with a ball of wool and hold it above ¼ teaspoon of tea leaves on your desk. Wait to see what happens.		

7	Rub a comb with a ball of wool and hold it next to an empty soft drink can which is on its side on a flat surface.		

- Rub a plastic comb or a plastic pen cover with a selection of materials.
- After rubbing each material hold the comb near some foam balls.
- Record the material you used and your observations.

No	Material	Observation (attracted or repelled the foam balls)
1		
2		
3		

Conclusion (What did I learn about static electricity?)